Proof by Induction: Inequalities and Non-standard Proofs

Questions 1–19

Question 1. Simplify each of the following:

a) 2×2^k

b) -3×3^k

c) $a \times a^k$

Question 2. Simplify each of the following:

a) $(1 + a) \times (1 + a)^k$

b) $(1 - a) \times (1 - a)^k$

Question 3. Simplify each of the following:

a) $3 \times 2!$

b) $12 \times 11!$

c) $(k+1)k!$

Question 4. Use proof by induction to show that $2^n > 2n$ for all positive integers n such that $n \geq 3$.

Question 5. Use mathematical induction to prove that $4^n > 3n + 7$ for all integers $n \geq 2$.

Question 6. Prove by induction, for all integers $n \geq 5$, that $n^2 < 2^n$.

Question 7. It is given that $n^3 < 3^n$ where n is a positive integer and a is the largest value of n for which the inequality does not hold true.

a) Find the value of a.

b) Show that it is true for $n < a$.

c) Prove, using mathematical induction, that for all $n > a$, $n^3 < 3^n$ holds true.

Question 8. Use proof by induction to show that $n! > 2^n$ for all positive integers n such that $n \geq 4$.

Question 9. Prove by induction, for all integers $n \geq 5$, that $n! > 10n$.

Question 10. Prove by induction, for all integers $n \geq 5$, that $3^n > (n+1)^3$.

Question 11. Prove by induction that $n! > n^2 + n$ for all integers $n \geq 4$.

Question 12. Using mathematical induction, prove that $2 > \left(1 + \frac{1}{n}\right)^3$ for all integers $n \geq 4$.

Question 13. For real number a and integer number n where $a > -1$ and $n \geq 2$, prove by mathematical induction that $(1 + a)^n > 1 + an$.

Question 14. Prove by induction, for all integers $n \geq 2$, that $n! < n^n$.

Question 15. Given that A and B are non-singular square matrices of equal dimensions. Use proof by induction to show that $(ABA^{-1})^n = AB^nA^{-1}$ for all integers $n \geq 1$.

Question 16. It is known that A is a square matrix such that $A^2 = A$. Use proof by induction to show that for all positive integers n, $I + A + A^2 + \cdots + A^n = I + nA$.

Question 17. It is known that A is a square matrix and a sequence of matrices is defined recursively as $A_1 = A$ and $A_{n+1} = A \times A_n$. Use proof by induction to show that for all positive integers n, $\det(A_n) = (\det(A))^n$.

Question 18. Use proof by induction to show that $\dfrac{d^n}{dx^n}(x^a) = \dfrac{a!}{(a-n)!}x^{a-n}$ for all integers $n \geq 1$ where $n \leq a$.

Question 19. Prove using mathematical induction that if there are n lines in a plane, no two of which are parallel and no three meet at a point, then they divide the plane into $\frac{n^2+n+2}{2}$ regions.